

Production of hydrocarbons by catalytic degradation of high density polyethylene in a laboratory fluidised-bed reactor

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Abstract

High density polyethylene (HDPE) was pyrolysed over different catalysts using an in-house designed laboratory fluidised-bed reactor operating in the 290–430°C range under atmospheric pressure. The catalysts used were HZSM-5, Silicalite, HMOR, HUSY and SAHA and the yield of volatile hydrocarbons (based on the feed) was typically HZSM-5>HUSY≈HMOR>SAHA. Product streams varied markedly with catalysts type and structure. The sodium form of siliceous ZSM-5, silicalite, containing very few or no catalytically active sites, gave very low conversions of polymer to volatile hydrocarbons. HZSM-5 and SAHA produced more olefinic streams with HZSM-5 yielding over 80 wt% olefins in the C₃–C₅ range compared with around 40 wt% with SAHA. However, both HMOR and HUSY produced more paraffinic streams with large amounts of C₄ (19 and 14 wt%, respectively) and both catalysts deactivated during the course of the degradation. Observed differences in product yields and product distributions under identical reaction conditions can be attributed to the microstructure of catalysts. © 1998 Elsevier Science B.V.

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1. Introduction

Polymer waste can cause serious pollution but also could be regarded as a cheap and abundant source of chemicals and energy. Although the calorific value of most plastic materials is very high, most waste poly-

mers are disposed of by landfill or incineration resulting in the loss of their energy content. Therefore, conversion of waste polymers to useful chemical feedstocks may be promising as a solution to their growing environmental problem and as a viable alternative to fossil fuels [1].

Possible technologies for the conversion of waste to useful products have attracted research in the area of thermal degradation. Workers in Japan have developed a dual fluidised-bed process for obtaining medium quality gases from municipal solid waste [2,3]. Thermal cracking of waste polymer using kilns or fluid beds has been piloted on a significant scale [4–7].

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However, the thermal degradation of polymers to low molecular weight materials has a major drawback in that a very broad product range is obtained. In addition, these processes require high temperatures, typically more than 500°C and even up to 900°C. Catalytic pyrolysis provides a means to address these problems. Suitable catalysts have the ability to control both the product yield and product distribution from polymer degradation as well as to significantly reduce the reaction temperature, potentially leading to a cheaper process with more valuable products.

Some polymeric materials, e.g. polystyrene, can be decomposed thermally in high yields to the monomers. However, this is not true for polyethylene or polypropylene which are among the most abundant polymeric waste materials typically making up 60–70% of municipal solid waste [8]. Their pyrolysis pattern is more complicated than in the case of polystyrene and it is difficult to obtain a high yield of ethylene or propylene. It would be desirable to convert these waste polyolefins into products of value other than the monomers, because the products could be of sufficient value to offset the collection and pyrolysis costs. Papers studying the catalytic process using adiabatic batch and fixed-bed reactors on the effect of the operations are numerous [9–14] dealing with experimental parameters including temperature, contact time, conversion and product distribution. However, the literature available involving the catalytic degradation of polyethylene at fluidised-bed reactor is rather scarce. A BP patent documents the use of a sand fluidised bed to crack polyolefins [15]. Another example was the attempt to use iron on activated carbon reacted at higher temperature (500–690°C) [16]. Our previous study [17] on the degradation of HDPE in a

fluidised bed noted that, when catalysts are used, degradation takes place at much lower temperatures, yielding large amounts of volatile products.

Potential concepts has been investigated using thermal analysis to estimate the amount of catalytic decomposition of the polymer and to predict kinetic behaviour [18,19]. Our study [20] has used thermogravimetric analysis (TGA) as a potential method for screening catalysts in the catalytic degradation of polyolefins. The objective of the research outlined in this paper is to expand considerably on previous work [17,20] using a laboratory catalytic fluidised-bed reactor, and specifically to study several concepts intended to achieve different product selectivities and enhance the potential benefit of catalytic polymer recycling.

2. Experimental

2.1. Materials and reaction preparation

The polymer used in this study was pure high density polyethylene (ex. BASF) with an average molecular weight of 75 000, as determined by gel permeation chromatography, and a density of 960 kgm⁻³. HDPE is a structurally regular chain material with very few small branch points (typically, less than 7 branch points per 1000 carbon atoms) and hence the polymer packs efficiently resulting in a highly crystalline material with correspondingly high density. The catalysts employed are described in Table 1. Prior to use, all the catalysts were pelleted using a press (compression pressure=160 MPa), crushed and sieved to give particle sizes ranging from

Table 1
Catalysts used in fluidised-bed reactor for HDPE degradation

Catalyst	Micropore size (nm)	BET area (cm ³ /g)	Si/Al ratio	Commercial name
HZSM-5	0.55×0.51	391	17	H-ZSM-5 zeolite ^a
HUSY	0.74	603	6.0	H-ultrastabilised Y zeolite ^b
HMOR	0.65×0.70	561	6.3	H-mordenite ^c
SAHA	3.15 ^d	274	2.6	Synclyst 25 ^b (Silica–alumina)

^a BP Chemicals, Sunbury-on-Thames, UK.

^b Crosfield Chemicals, Warrington, UK.

^c Laporte, Warrington, UK.

^d Single point BET determination.

75 to 180 μm . The catalyst (0.25–0.3 g) was then activated by heating, in the reactor, in flowing nitrogen (50 ml min^{-1}) to 120°C at 60°C h^{-1} . After 2 h, the temperature was increased to 520°C at a rate of 120°C h^{-1} . After 5 h, at 520°C , the reactor was cooled to the desired reaction temperature.

Several fluidisation runs were performed at ambient temperature and pressure to select suitable particle sizes (both catalyst and polymer) and to optimise the fluidising gas-flow rates to be used during the reaction. The particle size of both catalyst (75–180 μm) and polymer (75–250 μm) were chosen as being large enough to avoid entrainment. The catalyst were small enough to be adequately fluidised.

2.2. Experimental procedures and product analysis

A detailed description of the experimental system is given elsewhere [17] and shown schematically in Fig. 1. The reactor consists of a pyrex glass tube ($170 \times 20 \text{ mm i.d.}$, an upper section; $30 \times 10 \text{ mm i.d.}$, a middle section; $200 \times 10 \text{ mm i.d.}$, a lower section) with a sintered distributor (10 mm i.d., in the middle section) and a three-zone heating furnace with digital

controllers. High purity Nitrogen (BOC) was used as the fluidising gas and the flow was controlled by a needle valve. The polymer feed system was designed to allow HDPE particles, purged under nitrogen, to enter the top of the reactor and to drop freely into the fluidised bed at $t=0 \text{ min}$. On addition of the polymer, the fluidised bed remains fluidised [21]. The added polymer melts, wets the catalyst surface and is pulled into the catalyst macropores by capillary action. At sufficiently low polymer/catalyst ratios (as used here) the outside of the catalyst particles are not wet with polymer, so the catalyst particles move freely.

Volatile products leaving the reactor were passed through a glass-fibre filter to capture catalyst fines, followed by an ice–water condenser to collect any condensable liquid product. A three-way valve was used after the condenser to route product either into a sample gas bag or to an automated sample valve system with 16 loops. The Tedlar bags, 15 l capacity, were used to collect time-averaged gaseous samples. The bags were replaced at intervals of 10 min throughout the course of reaction. The multiport sampling valve allowed frequent, rapid sampling of the product stream at 0.5 and 1 min intervals. Gaseous products

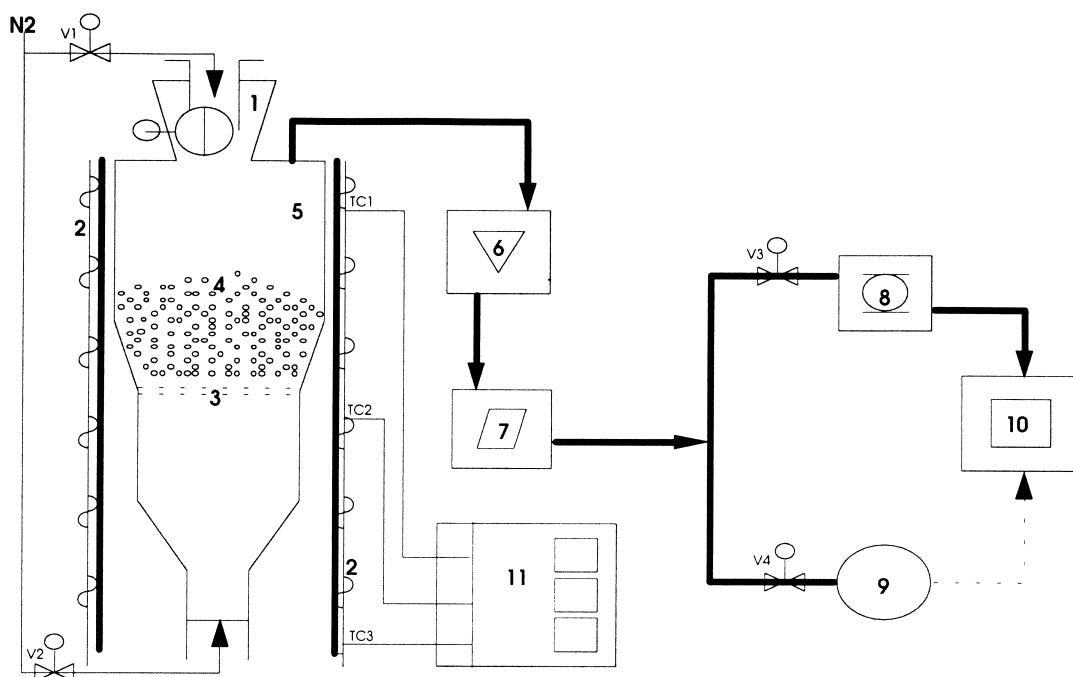


Fig. 1. Schematic diagram of catalytic fluidised-bed reactor system.

were analysed using a gas chromatograph (VARIAN 3400) equipped with (i) a thermal conductivity detector (TCD) fitted with a 1.5 m×0.2 mm i.d. Molecular Sieve 13X packed column and (ii) a flame ionisation detector (FID) fitted with a 50 m×0.32 mm i.d. PLOT Al₂O₃/KCl capillary column. A calibration cylinder containing 1% C₁–C₅ hydrocarbons (Linde Gas, UK) was used to help identify and quantify the gaseous products.

The remaining solids deposited on the catalyst after the catalytic degradation of the polymer were deemed 'residues' and contained involatile products and coke. The amount and nature of the residues were determined by thermogravimetric analysis (TA Instruments, SDT 2960 Simultaneous DTA–TGA) as described elsewhere [22].

3. Results and discussion

Catalytic pyrolysis products (*P*) are grouped together as hydrocarbon gases (<C₅), gasoline (C₅–C₉) and residues to enable the overall pyrolysis

processes to be described more easily. The term 'yield' as used in this paper is defined by the relation:

$$\text{Yield}(\text{wt}\%) = (P(\text{g}) \times 100) / \text{Polymer fed}(\text{g})$$

3.1. Degradation of HDPE over SAHA, HZSM-5 and HUSY

Similar trends in product yields were observed with all catalysts as the reaction temperature was increased. Gas and coke yields increased and involatile residues (unreacted or partially reacted HDPE) and liquids decreased (Tables 2–4).

Much higher involatile residues were observed when using SAHA as the catalyst compared with HZSM-5 and HUSY. This can easily be explained in terms of the nature of the acid sites with SAHA being made up predominantly of weaker Lewis sites compared to the predominantly more acidic Brønsted sites found in zeolite catalysts.

The rate of gaseous hydrocarbon evolution further highlights the slower rate of degradation over SAHA as shown in Fig. 2 when comparing all catalysts under

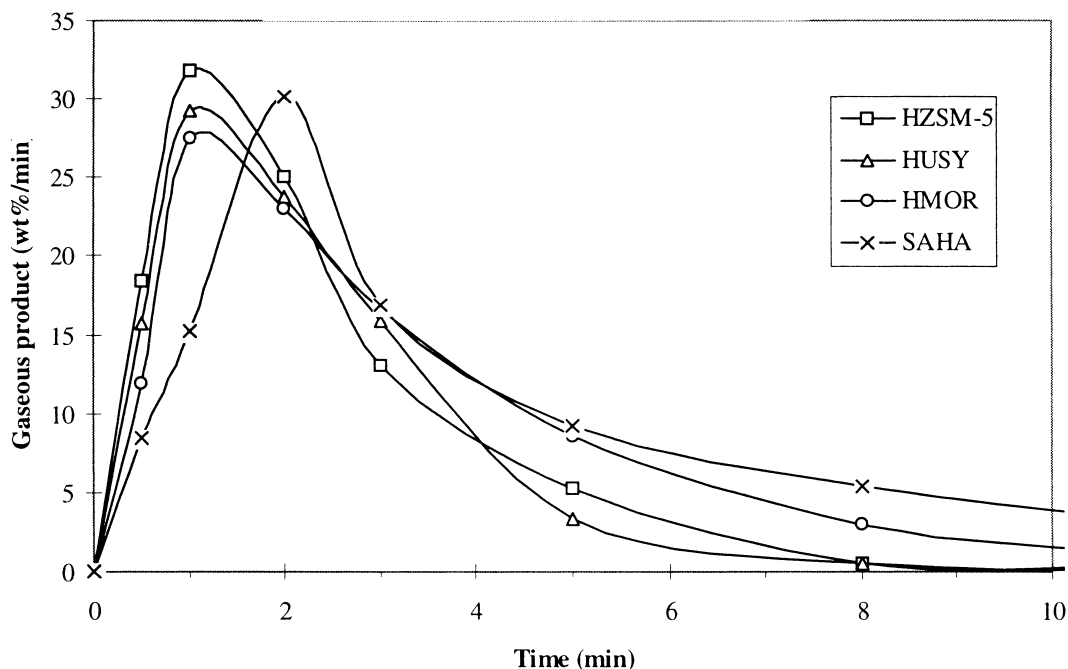


Fig. 2. Comparison of hydrocarbon yields as a function of time for the catalytic pyrolysis over different catalysts (polymer-to-catalyst ratio=40% wt/wt, rate of fluidisation gas=570 ml/min, reaction temperature=360°C).

Table 2

Product distributions for HZSM-5 catalysed degradation of HDPE at different reaction temperatures (fluidising N₂ rate=570 ml min⁻¹, polymer-to-catalyst ratio=40% wt/wt, total time of collection=30 min)

Degradation results	Temperature (°C)				
	290	330	360	390	430
Yield (wt% feed)					
Gaseous	86.99	90.11	93.23	95.20	95.38
Liquid ^a	5.19	4.09	2.27	1.44	1.61
Residue	7.82	4.82	4.53	3.36	3.03
Involatile residue	6.23	3.17	2.82	1.63	1.30
Coke	1.59	1.65	1.71	1.73	1.73
Mass balance (%)	92.10	89.06	90.32	93.68	94.18
Distribution of gaseous products (wt% feed)					
Hydrocarbon gases ($\sum C_1-C_4$)	55.09	58.21	64.14	64.85	65.50
C ₁	n.d. ^b	n.d. ^b	n.d. ^b	0.01	0.02
C ₂	n.d. ^b	— ^c	— ^c	0.05	0.07
C ₂ =	0.79	1.86	2.65	4.15	4.91
C ₃	0.86	1.93	3.78	4.02	4.24
C ₃ =	19.47	20.11	20.82	22.06	21.65
C ₄	4.97	10.64	10.21	10.59	11.00
C ₄ =	28.99	23.66	26.67	23.95	23.60
Gasoline ($\sum C_5-C_9$)	31.42	30.49	27.09	26.04	24.80
C ₅	2.93	5.92	5.02	4.68	4.83
C ₅ =	17.67	13.68	12.50	11.44	10.37
C ₆	2.28	2.34	2.05	2.55	2.46
C ₆ =	5.30	4.31	4.18	4.01	3.60
C ₇	1.12	2.36	1.76	1.52	1.91
C ₇ =	1.77	1.11	1.07	0.93	1.22
C ₈	0.30	0.72	0.37	0.76	0.37
C ₈ =	0.03	0.04	0.11	0.13	0.04
$\sum C_9$	0.02	0.01	0.03	0.01	— ^c
BTX ^d	0.48	1.41	2.00	4.31	5.09

^a Condensate in condenser and captured in filter, unidentified.

^b Not detectable.

^c Less than 0.01 (wt%).

^d Benzene, toluene and xylene.

identical conditions at 360°C. The maximum rate of generation was observed after 1 min with the zeolite catalysts whereas the maximum was observed after 2 min with SAHA.

Overall, the change in the hydrocarbon yield with temperature was similar for all catalysts with faster rates observed at higher temperatures (Fig. 3 shows typical variations with temperature for HUSY). As the temperature of reaction was decreased, the initial rate of hydrocarbon production dropped and the time for HDPE to be degraded lengthened. At 290°C, the rate of hydrocarbon production was significantly lower

with the polymer being degraded more slowly over 15 min.

3.2. Product stream variation with catalyst

Paraffin cracking is well known to proceed over zeolites and silica–aluminas by carbocation mechanisms, where the initially formed ions undergo chain reactions via processes, such as isomerisation and β -scission (monomolecular) and hydrogen transfer and oligomerisation (bimolecular), to yield typically C₃–C₆ paraffins and olefins [23–25]. Both the carbon number distribution of the products of HDPE cracking

Table 3

Product distributions for HUSY catalysed degradation of HDPE at different reaction temperatures (fluidising N₂ rate=570 ml min⁻¹, polymer-to-catalyst ratio=40 wt%/wt%, total time of collection=30 min)

Degradation results	Temperature (°C)				
	290	330	360	390	430
Yield (wt% feed)					
Gaseous	84.70	89.68	89.71	91.82	93.40
Liquid ^a	5.90	3.38	3.22	2.68	1.43
Residue	9.40	6.94	7.07	5.50	5.17
Involatile residue	5.74	3.12	3.13	1.47	1.06
Coke	3.66	3.82	3.94	4.03	4.11
Mass balance (%)	83.11	86.41	85.82	91.77	91.04
Distribution of gaseous products (wt% feed)					
Hydrocarbon gases ($\sum C_1-C_4$)	25.83	33.71	34.81	37.56	40.07
C ₁	n.d. ^b	n.d. ^b	— ^c	0.03	0.24
C ₂	n.d. ^b	— ^c	0.03	0.07	0.13
C ₂ ⁼	0.09	0.19	0.29	0.38	0.56
C ₃	0.23	0.39	0.58	0.68	0.99
C ₃ ⁼	5.72	8.35	8.75	9.64	10.35
C ₄	12.88	13.43	13.82	15.06	15.37
C ₄ ⁼	6.91	11.35	11.33	11.70	12.43
Gasoline ($\sum C_5-C_9$)	57.28	54.91	53.67	52.85	51.62
C ₅	15.76	14.61	14.45	15.94	15.07
C ₅ ⁼	5.26	8.50	7.52	7.41	8.60
C ₆	14.03	12.15	13.01	12.83	11.48
C ₆ ⁼	1.71	4.06	4.26	2.89	5.16
C ₇	10.17	8.84	7.49	7.46	6.35
C ₇ ⁼	0.20	0.21	1.05	0.40	1.02
C ₈	6.76	4.07	3.95	3.87	2.92
C ₈ ⁼	2.26	1.60	1.21	1.09	0.58
$\sum C_9$	1.14	0.88	0.74	0.96	0.44
BTX ^d	1.59	1.06	1.23	1.40	1.71

^a Condensate in condenser and captured in filter, unidentified.

^b Not detectable.

^c Less than 0.01 (wt%).

^d Benzene, toluene and xylene.

at 360°C over the various catalysts used in this study (Table 5) and the nature of the product distribution were found to vary with the catalyst used. As expected, the conversion of HDPE over silicalite at 360°C was only 4.1 wt% compared with 93.2 wt% when HZSM-5 was used. Typically the product stream was C₃–C₆ olefins with some isomerisation of C₄ and above. At higher temperatures, predominantly thermally cracked products were observed [17], with product streams containing C₁–C₉ hydrocarbons and 82 wt% of HDPE converted at 520°C. SAHA with large mesopores and low acidity gave rise to the broadest carbon range of C₃–C₈. The weaker Lewis-acid sites

of the SAHA catalyst resulted in a highly olefinic product ($\sum \text{olefin} / \sum \text{paraffin}(\text{o/p}) = 8.6$, see Table 6) and the largest amount of unreacted feed. By comparison, the stronger acid sites of the zeolite catalysts resulted in increased bimolecular hydrogen transfer in the order:

$$\text{HZSM-5}(\text{o/p} = 2.9) < \text{HMOR}(\text{o/p} = 1.5) \\ < \text{HUSY}(\text{o/p} = 0.6)$$

Bimolecular reactions, such as hydrogen transfer, are sterically hindered within the 10-ring channel system (approximate diameter of 0.55×0.51 nm) of HZSM-5. However, the 12-ring channels of mordenite

Table 4

Product distributions for SAHA catalysed degradation of HDPE at different reaction temperatures (fluidising N₂ rate=570 ml min⁻¹, polymer-to-catalyst ratio=40% wt/wt, total time of collection=30 min)

Degradation results	Temperature (°C)				
	290	330	360	390	430
Yield (wt% feed)					
Gaseous	76.29	81.00	84.47	85.46	88.56
Liquid ^a	6.21	4.79	3.40	2.58	2.51
Residue	17.50	14.21	12.13	11.96	8.93
Involatile residue	15.02	11.77	9.62	9.34	6.12
Coke	2.48	2.44	2.51	2.62	2.81
Mass balance (%)	84.46	86.74	88.24	90.61	88.52
Distribution of gaseous products (wt% feed)					
Hydrocarbon gases ($\sum C_1-C_4$)	28.65	29.87	30.26	29.49	29.98
C ₁	n.d. ^b	n.d. ^b	n.d. ^b	n.d. ^b	n.d. ^b
C ₂	n.d. ^b	n.d. ^b	n.d. ^b	n.d. ^b	n.d. ^b
C ₂ ⁼	— ^c	— ^c	0.01	0.01	0.02
C ₃	0.01	0.02	0.03	0.11	0.15
C ₃ ⁼	7.56	7.75	7.62	7.39	7.95
C ₄	1.83	1.78	1.71	1.53	1.89
C ₄ ⁼	19.26	20.31	20.88	20.45	19.97
Gasoline ($\sum C_5-C_9$)	47.64	51.13	54.21	55.96	58.58
C ₅	1.13	1.41	1.41	1.30	1.63
C ₅ ⁼	22.12	23.47	23.27	24.11	22.12
C ₆	1.63	2.34	2.10	2.17	2.82
C ₆ ⁼	14.70	15.11	15.95	15.45	16.72
C ₇	2.02	2.49	3.07	2.88	3.31
C ₇ ⁼	5.75	5.89	7.18	8.80	9.86
C ₈	0.13	0.24	0.45	0.51	1.17
C ₈ ⁼	0.15	0.18	0.58	0.47	0.90
$\sum C_9$	0.01	— ^c	0.21	0.28	0.05
BTX ^d	n.d. ^b	n.d. ^b	— ^c	0.01	0.01

^a Liquid: condensate in condenser and captured in filter, unidentified.

^b Not detectable.

^c Less than 0.01 (wt%).

^d BTX: benzene, toluene and xylene.

(0.65×0.70 nm) are less restrictive; hence the increase in the paraffin product. The combination of 12-ring pore openings and large internal supercages of HUSY allow significant bimolecular reactions and yielded a saturate-rich product with a wide carbon number distribution and substantial coke levels.

HZSM-5 and HMOR, both containing more restricted channel systems, yielded a narrow product distribution with over 80 wt% olefins in the C₃–C₅ range. HMOR was particularly selective to C₄ products generating over 44 wt% at 360°C with the highest yield of *i*-C₄. The *i*-C₄/*n*-C₄ ratio reflects the involvement of tertiary C₄ carbenium ions in bimolecular hydrogen transfer reactions. Since tertiary

carbenium ions are more stable than primary ions, a higher yield of *i*-C₄ would be expected. The observed *i*-C₄/*n*-C₄ ratios at 360°C are well above calculated equilibrium values and are consistent with the steric constraints of their structures following the order:

ZSM-5(*i*-C₄/*n*-C₄ = 4.6)

< mordenite(*i*-C₄/*n*-C₄ = 6.7)

< US-Y(*i*-C₄/*n*-C₄ = 39.1)

The high ratio is consistent with the cracking of branched oligomers to yield *i*-C₄. Much higher *i*-C₄/*n*-C₄ ratios have been observed in hexane cracking for large pore 3D zeolites, such as Beta and Y, compared

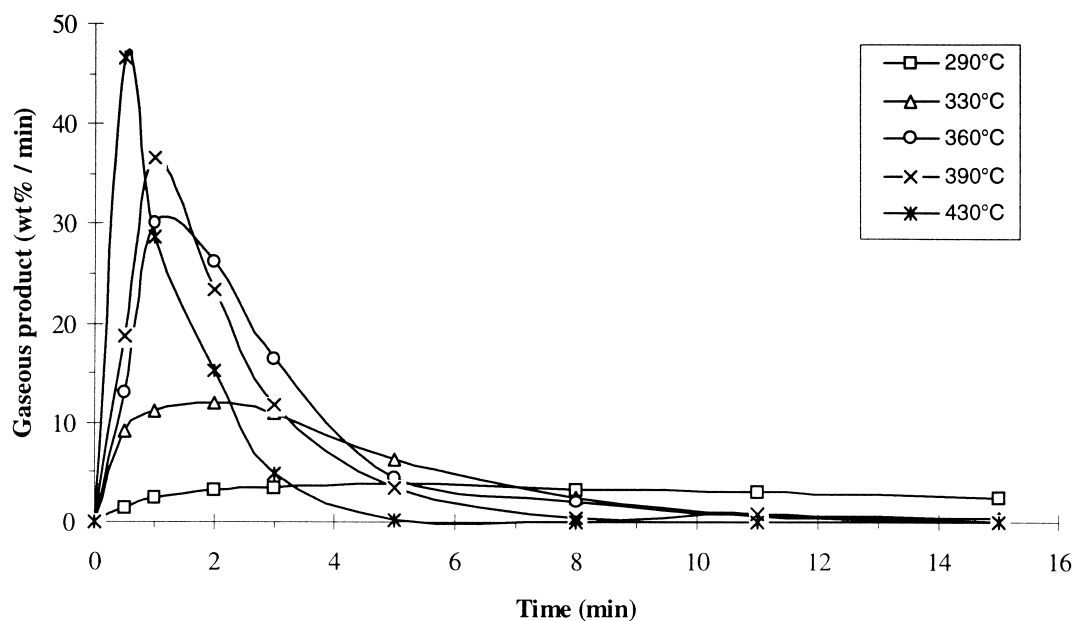


Fig. 3. Comparison of hydrocarbon yields as a function of time at different reaction temperatures for the catalytic degradation of HDPE over HUSY (rate of fluidising gas=570 ml/min, ratio of polymer to catalyst=40% wt/wt).

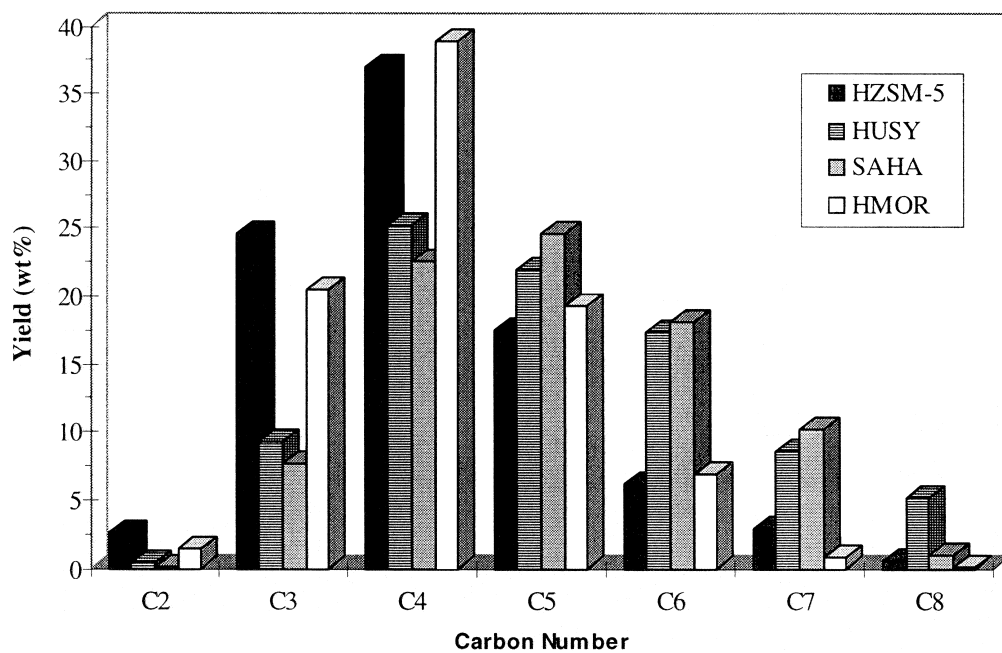


Fig. 4. Comparison of hydrocarbon yields for the catalytic degradation of HDPE over various catalysts (polymer-to-catalyst ratio=40% wt/wt, rate of fluidisation gas=570 ml/min, reaction temperature=360°C).

Table 5

Product distributions for the degradation of HDPE over different catalysts (reaction temperature=360°C, fluidising N₂ rate=570 ml min⁻¹, polymer-to-catalyst ratio=40 wt%/wt%, total time of collection=30 min)

Degradation results	Catalyst type				
	ZSM-5	USY	SAHA	MORD	Silicalite ^a
Yield (wt% feed)					
Gaseous	93.23	89.71	84.47	88.56	4.1
Liquid ^b	2.27	3.22	3.40	2.51	1.5
Residue	4.53	7.07	12.13	8.93	94.4
Involatile residue	2.82	3.13	9.62	6.12	94.4
Coke	1.71	3.94	2.51	2.81	— ^c
Mass balance (%)	90.32	85.82	88.24	88.52	98.94
Distribution of gaseous products (wt% feed)					
Hydrocarbon gases ($\sum C_1-C_4$)	64.14	34.81	30.26	60.77	3.2
C ₁	n.d. ^d	— ^c	n.d. ^d	n.d. ^d	n.d. ^d
C ₂	— ^c	0.03	n.d. ^d	— ^c	n.d. ^d
C ₂ ⁼	2.65	0.29	0.01	1.36	n.d. ^d
C ₃	3.78	0.58	0.03	3.89	n.d. ^d
C ₃ ⁼	20.82	8.75	7.62	16.56	1.3
C ₄	10.21	13.82	1.71	19.29	n.d. ^d
C ₄ ⁼	26.67	11.33	20.88	19.67	1.9
Gasoline ($\sum C_5-C_9$)	27.09	53.67	54.21	27.79	0.9
C ₅	5.02	14.45	1.41	7.87	n.d. ^d
C ₅ ⁼	12.50	7.52	23.27	11.42	0.9
C ₆	2.05	13.01	2.10	3.37	n.d. ^d
C ₆ ⁼	4.18	4.26	15.95	3.55	0.01
C ₇	1.76	7.49	3.07	0.50	n.d. ^d
C ₇ ⁼	1.07	1.05	7.18	0.27	n.d. ^d
C ₈	0.37	3.95	0.45	— ^c	n.d. ^d
C ₈ ⁼	0.11	1.21	0.58	— ^c	n.d. ^d
$\sum C_9$	0.03	0.74	0.21	0.77	n.d. ^d
BTX ^c	2.00	1.23	— ^c	0.03	n.d. ^d

^a Sodium form of siliceous ZSM-5 containing very few or no catalytically active sites [17].

^b Liquid: condensate in condenser and captured in filter, unidentified.

^c Less than 0.01 (wt%).

^d Not detectable.

^e BTX: benzene, toluene and xylene.

Table 6

Influence of catalyst types and reaction temperatures on product selectivity for the degradation of HDPE – Experimental and predicted equilibrium results

Ratio	Catalyst type ^a (°C)												VEL ^b (°C)		
	HZSM-5			HUSY			SAHA			HMOR			290	360	430
	290	360	430	290	360	430	290	360	430	290	360	430			
<i>i</i> -C ₄ / <i>n</i> -C ₄	4.9	4.6	3.2	39.1	20	12.4	43.2	24.5	18.9	— ^c	6.7	— ^c	1.17	0.95	0.81
o/p ^d	5.9	2.9	2.6	0.4	0.6	0.7	10.4	8.6	7.1	— ^c	1.5	— ^c	— ^c	— ^c	— ^c

^a Represents a series of runs where 570 ml min⁻¹ N₂ fluidising rate and 40% wt/wt polymer to catalyst feed.

^b Predicted equilibrium data.

^c Data not available.

^d Denotes the ratio of the sum of all olefinic to paraffin products.

to HZSM-5, because in the absence of the steric constraints of the HZSM-5 structure the formation of bulky bimolecular reaction intermediates is not restricted [26].

The yield of smaller cracked products increased with temperature as did the yields of BTX and coke. Further evidence of the increase in secondary reactions was seen in the lowering of the $i\text{-C}_4/n\text{-C}_4$ and o/p ratios with temperature (Table 6). The pore structure of HZSM-5 restricts the formation of bulky intermediates and, consequently, the catalyst is resistant to coke

formation explaining the relatively low values observed at high conversions and the very small increase with increasing reaction temperature (Table 2).

3.3. Variation in the product stream with time

Rapid deactivation of both HMOR and HUSY was observed (Figs. 4 and 5) when the spot samples, taken during the course of the reaction, were analysed. The deactivation is reflected in the decrease of the amount

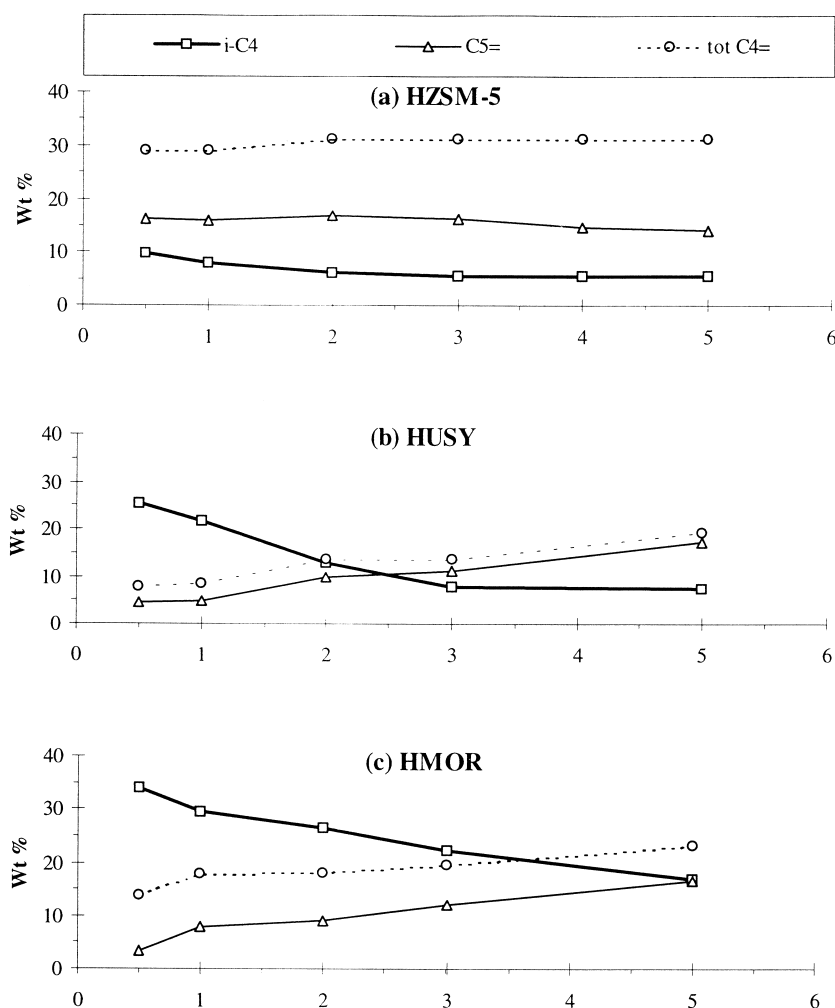


Fig. 5. Comparison of some of the main hydrocarbon products ($i\text{-C}_4$, $\text{C}_5=$ and total $\text{C}_4=$) as a function of time for the catalytic degradation of HDPE over (a) HZSM-5, (b) HUSY, (c) HMOR and (d) SAHA catalysts (rate of fluidising gas=570 ml/min, ratio of polymer to catalyst=40% wt/wt).

of *i*-C₄ produced (product of bimolecular reaction) and the relative increase in olefins (product of monomolecular reaction), exemplified by, C₄ and C₅. As mentioned earlier, the larger 12-ring pore openings and channel system of HMOR allow bulky bimolecular reactions to occur, ultimately leading to the generation of coke and, subsequently, deactivation of the catalyst. The deactivation was more exaggerated in the case of HUSY with its large pore openings and internal supercages. In contrast, HZSM-5 and SAHA product streams remain virtually unchanged throughout the degradation of HDPE. As previously stated, HZSM-5 is resistant to coking and the product stream remains especially unchanged, whereas the weakness of the acid sites in SAHA is most likely the reason for the lack of variation in the product stream over this catalyst.

4. Conclusions

A laboratory-scale catalytic fluidised-bed reactor has been used to obtain a range of volatile hydrocarbons by catalytic pyrolysis of HDPE polymer in the 290–430°C range. The fluidised bed has a number of special advantages for the pyrolysis of HDPE; it is characterised by excellent heat and mass transfer, it has a low tendency for clogging with molten polymer and has the ability to maintain a nearly constant temperature throughout the reactor.

The results from this work confirm that although the initial cracking of HDPE must be confined to the external surface and pore mouths of the zeolite catalysts, the resultant initial cracked products are then degraded further within the zeolite. Clearly, both acidity and diffusion constraints within individual micropores of each zeolite may play significant roles in the observed product distribution from HDPE cracking. The acidic zeolite catalysts, HZSM-5, HMOR and HUSY, are more effective in converting the polymer to volatile hydrocarbons than the less acidic amorphous SAHA.

SAHA showed the lowest conversion and generated an olefin-rich product, predominantly between C₃–C₈, whereas HUSY yielded a saturate-rich product with a wide carbon number distribution and substantial coke levels. Greater product selectivity was observed with HZSM-5 and HMOR as catalysts with over 80% of the

product in the C₃–C₅ range and HMOR generating the highest yield of *i*-C₄ for all catalysts studied. Both the larger pore zeolites (HUSY and HMOR) showed deactivation in contrast to the more restrictive HZSM-5. The less acidic catalyst, SAHA, was less reactive and did not appear to deactivate under the reaction conditions.

The systematic experiments discussed in this paper and in earlier work [17] show that the use of a catalyst significantly reduces the required reaction temperature and improves the yield of volatile products as well as providing better selectivity in the product distributions. The catalytic degradation of polyethylene performed in fluidised-bed reactor was thus found to be a useful method for the production of potentially valuable hydrocarbons.

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